| | | | |
|---|---|---|---|
| Prepared By | 08-Aug-2012 | Mr.Chatchawan S. | RNC<br>RENEWCOILS ENGINEER & SUPPLY CO.,LTD. |
| Checked By | 08-Aug-2012 | Mr.Chatchawan S. | |
| Approved By | 08-Aug-2012 | Mr.Amnauy W. | |
| Certified By | 08-Aug-2012 | Mr.Amnauy W. | Renew coils Engineering & Supply Co., Ltd. |

| Rev. | Date | Checked | Approved | Description |
|---|---|---|---|---|
| 0 | 08-Aug-2012 | Mr.Chatchawan S. | Mr.Amnauy W. | Issue for Approved |

## Renew Coils Engineering & Supply Co., Ltd.

## Doc. No. : RNC-WPS-018-2012

## Addendum

## WELDING PROCEDURE SPECIFICATION (WPS)
## RNC-WPS-008

# WELDING PROCEDURE SPECIFICATION (WPS)

RENEWCOILS ENGINEER & SUPPLY CO.,LTD.

Company Name: **Renew Coil Engineering & Supply Co., Ltd.**

WPS No: RNC-WPS-008 | Date: 8-Aug-12 | Revision No: 0

Supporting PQR No.(s): RNC-PQR-008 | Date: 8-Aug-12 | Type(s): Manual

Welding Process(es): Gas Tungsten Arc Welding (GTAW)

## JOINT (QW-402)

Joint Design: **Single V**

Backing (Yes) ______ (No) X

Backing Material (Type): N/A

☐ Metal ☐ Nonfusing Metal
☐ Nonmetallic ☐ Other

Notes: ______

**See Details Attachment sheet Joint Sketch**

## BASE METALS (QW-403)

P-No: 8 | Group No: 1 | to P-No: 8 | Group No: 1

Material Specification Type and Grade: A 312 Gr.TP316L, A182 F316L TO A 312 Gr.TP 316L, A182 F316L

Chem. Analysis and Mech. Prop: ______

Thickness Range

Base Metal: Groove 1.6 mm - 14.22 mm | Fillet All

Pipe Dia. Range: Groove A.I | Fillet All

Other: ______

## FILLER METALS (QW-404)

| | GTAW | SMAW |
|---|---|---|
| Spec. No. (SFA) | 5.9 | 5.4 |
| AWS No. (Class) | ER-316L | E-316L-16 |
| F-No | 6 | 5 |
| A-No. | 8 | 8 |
| Size of Filler Metals | Ø 2.0 & 2.4 mm | Ø 3.2 mm |
| Deposited Weld Metals | 2.0 mm | 5.11 mm. |
| **Thickness Range :** | | |
| Groove | 4.0 mm Max | 10.22 mm |
| Fillet | All | All |
| Electrode-Flux (Class) | - | - |
| Flux Trade Name | - | - |
| Consumable Insert | - | - |
| Other | Kobelco TGS-316L or Equivalents | Kobelco NC-36L or Equivalents |

## POSITION (QW-405)

Position(s) of Groove: ALL

Welding Progression: Up X Down ______

Position(s) of Fillet: ALL

## POSTWELD HEAT TREATMENT (QW-407)

Temperature Range: N/A

Time Range: N/A

Notes: ______

## PREHEAT (QW-406)

Preheat Temp. Min.: 10°C

Interpass Temp. Max: 150°C

Preheat Maintenance: N/A

Notes: ______

## GAS (QW-408)

Percent Composition

| | Gas(es) | (Mixture) | Flow Rate |
|---|---|---|---|
| Shielding | Argon | 99.99% | 12-16 Ltr/Min |
| Trailing | - | - | - |
| Backing | Argon | 99.99% | 13-17 Ltr/Min |

**Sheet 1 of 2**

RNC
RENEWCOILS ENGINEER & SUPPLY CO.,LTD.

# WELDING PROCEDURE SPECIFICATION (WPS)

WPS No RNC-WPS-008 Rev. 0

## ELECTRICAL CHARACTERISTICS (QW-409)

Current [X] AC [X] DC Polarity EN(GTAW), EP(SMAW)

Amps (Range) 80-110(GTAW) 80-130(SMAW) Volts (Range) 8-13(GTAW), 18-24(SMAW)

Tungsten Electrode Size and Type Ø 2.4 mm Thoriated ( EWTh-2 )

(Pure Tungsten, 2% Thoriated, etc.)

Mode of Metal Transfer for GMAW N/A

(Spray arc, short circuiting arc, etc.)

Electrode Wire feed speed range N/A

## TECHNIQUE (QW-410)

String or Weave Bead BOTH APPLICATIONS

Orifice or Gas Cup Size 8.0 mm - 19.0 mm

Initial and Interpass Cleaning (Brushing, Grinding, etc.) BRUSHING OR GRINDING

Method of Back Gouging N/A

Oscillation N/A

Contact Tube to Work Distance N/A

Multiple or Single Pass (per side) MULTIPASS

Multiple or Single Electrodes SINGLE ELECTRODE

Travel Speed (Range) 5-10 Cm/Minute

Peening N/A

Other N/A

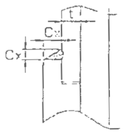
| Weld Layer(s) | Process | Filler Metal Class | Filler Metal Dia (mm) | Current Type Polar | Current Amp Range | Volt Range | Travel Speed Range (Cm/min) | Other |
|---|---|---|---|---|---|---|---|---|
| 1st | GTAW | ER-316L | 2.4 | DC EN | 80-110 | 8-12 | 5-9 | |
| 2nd | GTAW | ER-316L | 2.4 | DC EN | 100-130 | 10-13 | 7-11 | |
| 3 & Over | SMAW | E-316L-16 | 2.6-3.2 | AC DCEP | 80-110 | 18-24 | 6.0-10.0 | |
| | | | | | | | | |
| | | | | | | | | |
| | | | | | | | | |
| | | | | | | | | |

We, the undersigned, certify that the statements in this record are correct and that the test welds were prepared, welded and tested in accordance with the requirement of **ASME-IX, 2007 latest Revision.**

| | Name | Signed | Date |
|---|---|---|---|
| Prepared by | Mr. Chatchawan S. | [signature] | 18/08/12 |
| Reviewed by : | Mr. Chatchawan S | [signature] | 18/08/12. |
| Approved by: | AMNUAY W. | [signature] | 20-08-55 |

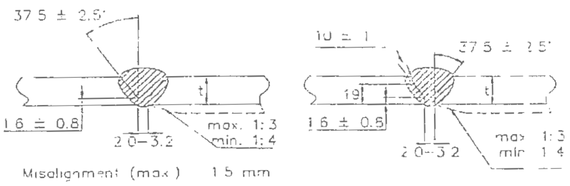
**Sheet 2 of 2**

# ATTACHED FIGURE JOINT SKETCH



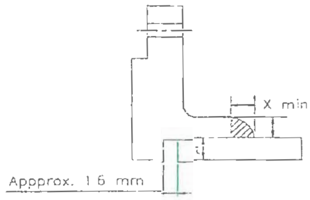

Cx (min.)=1.25t but not less than 3.2 mm

JOINT No. A1



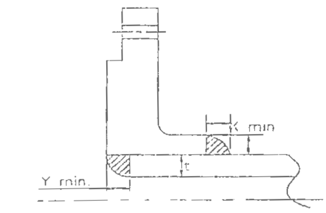

Misalignment (max.) 1.5 mm

JOINT No. B1 & B2



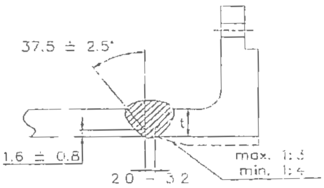

X min. = The lesser of 1.4t or thickness of the hub.

JOINT No. A2



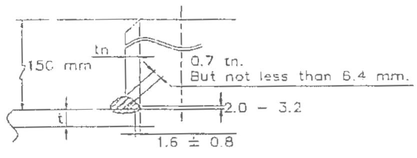

X min. = The lesser of 1.4t or thickness of the hub.
Y min. = The lesser of t or 6.4 mm.

JOINT No. A3



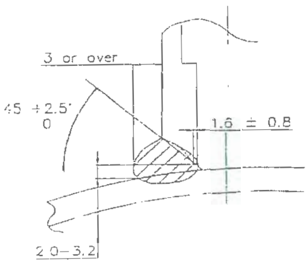

Misalignment (max.) = 1.5 mm

JOINT No. B3

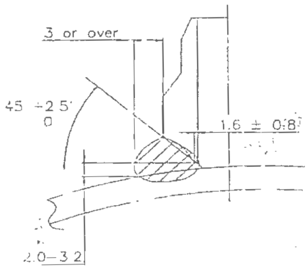



Misalignment (max.) = The lesser of 3.2 mm or 0.5tn.

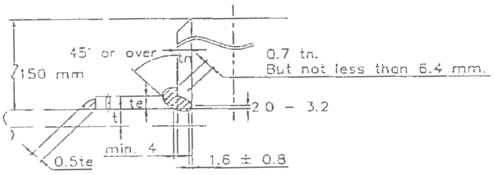
JOINT No. B4




JOINT No. C1

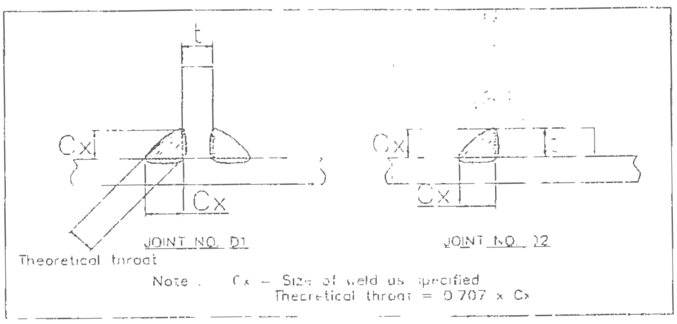



JOINT No. C2




Misalignment (max.) = The lesser of 3.2 mm or 0.5tn.

JOINT No. B5



# Procedure Qualification Record (PQR)

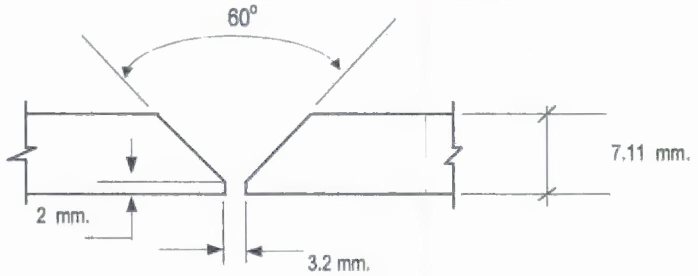
ASME-QW-200.2, Section IX, ASME Boiler Pressure Vessel Code

Record Actual Variables Used to Weld Test Coupon

## RECORD OF WELDING (QW-483) PQR Number : RNC-PQR-008

| | | | |
|---|---|---|---|
| Company Name : | RENEWCOILS ENGINEERING & SUPPLY CO.,LTD. | | |
| PQR Record Number : | RNC-PQR-008 | Date : | 6 August 2012 |
| WPS Number : | RNC-WPS-008 | | |
| Welding Process (es) : | Gas Tungsten Arc Welding (GTAW) + Shield Metal Arc Welding ( SMAW ) | | |
| Type (Manual, Semi-auto, or Automatic ): | Manual | | |

## JOINT DESIGN (QW-402)




(For combination qualifications, the deposited weld metal thickness shall be recorded for each filler metal or process used)

## WELDING VARIABLES

### BASE METALS (QW-403)

| | | | |
|---|---|---|---|
| Material Spec. | A312 | To | A312 |
| Type or Grade | TP316L | To | TP316L |
| P. No. | P.8 | To | P.8 |
| Thickness of Test Coupon | 7.11 mm. | | |
| Diameter of Test Coupon | Ø 6" | | |
| Others | | | |

### FILLER METALS (QW-404)

| | GTAW | SMAW |
|---|---|---|
| SFA Specification | A 5.9 | A 5.4 |
| AWS Classification | ER316L | E316L-16 |
| Filler Metal F-No. | 6 | 5 |
| Weld Metal Analysis A-No. | 8 | 8 |
| Size of Filler Metal | Ø 2.4 mm. | Ø 3.2 mm. |
| Others | Kobelco TGS-316L | Kebelco NC-36L |
| Weld Metal Thickness | - | |

### POSITION (QW-405)

| | |
|---|---|
| Position of Groove | 6G |
| Progression of Welding (Uphill or Down) | Up Hill |
| Others | N/A |

### PREHEAT (QW-406)

| | |
|---|---|
| Preheat Temperature | Ambient Temperature |
| Interpass Temperature (max.) | 140° C |
| Others | N/A |

### POSTWELD HEAT TREATMENT (QW-407)

| | |
|---|---|
| Temperature | N/A |
| Hold Time | N/A |
| Others | |

### GAS (QW-408)

| | Percent Composition | | |
|---|---|---|---|
| | Gas (es) | Mixture | Flow Rate |
| Shielding | Argon | 99.95% | 20 L/Min. |
| Trailing | - | - | - |
| Backing | Argon | 99.95% | 20 L/Min. |

### ELECTRICAL CHARACTERISTICS (QW-409)

| | | | |
|---|---|---|---|
| Current | DC (GTAW) | | DC (SMAW) |
| Polarity | EN | | EP |
| Amperage | 82-100 for GTAW and 119 - 122 for SMAW | Voltage | 11-12 for GTAW 25 for SMAW |
| Tungsten Electrode Size | ø 2.4 mm. 2% thoriated.EWTh-2 | | |
| Others | N/A | | |

### TECHNIQUE (QW-410)

| | |
|---|---|
| Travel Speed | 3.82 - 9.20 cm/min. |
| String or Weave Bead | Both |
| Oscillation | N/A |
| Multipass or single Pass | Multiple Pass |
| Multipass or Single Electrodes | Multiple Pass |
| Others | N/A |



*PAE Technical Service Company Limited*

Public Registration No. (5) 1571/2542

69 Srinakarin Rd., suanluang, Bangkok 10250, Thailand. Tel (662) 322-022 Fax : (622) 721-2577

# Procedure Qualification Record

## ASME-QW-200.2, Section IX (WPQR)

RECORD OF TEST RESULTS (QW-483) PQR Number-: RNC-PQR-008

### Tensile Test Report No.TS-12-08-079

| Specimen | Width (mm) | Thickness (mm) | Area ($mm^2$) | Ultimate Total Load (KN) | Ultimate Strength ($N/mm^2$) | Type of Failure & Loc. |
|---|---|---|---|---|---|---|
| 3178/12TR1 | 19.51 | 6.58 | 128.376 | 68.803 | 535.949 | OUT OF WELD |
| | | | | | | |
| 3178/12 TR2 | 19.46 | 6.45 | 125.517 | 69.196 | 551.288 | OUT OF WELD |

### Guided Bend Tests (QW-160) Report No.BD-12-08-018

| Type and Figure Number | Type of Bend | Indication | Results |
|---|---|---|---|
| 3178/12 BF1 | Transverse face bend | No Open Discontinuity | Accepted |
| 3178/12 BF2 | Transverse face bend | No Open Discontinuity | Accepted |
| 3178/12 BR1 | Transverse root bend | No Open Discontinuity | Accepted |
| 3178/12 BR2 | Transverse root bend | No Open Discontinuity | Accepted |
| | | | |

### Toughness Tests (QW-170)

| Specimen Number | Notch Location | Notch Type | Test Temp. | Impact Values | Lateral Expansion | | Drop Weight | |
|---|---|---|---|---|---|---|---|---|
| | | | | | % Shear | Mils | Break | No Break |
| | | | | | | | | |
| N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A |
| | | | | | | | | |
| | | | | | | | | |

### Fillet Weld Test (QW-180)

Result - acceptable : Yes N/A No N/A Penetration Into Base Metal : Yes N/A No N/A

Macroetch Test Results N/A

### Other Tests

Type of Test Radiographic examination results Accepted Report No.PQR - 012

Hardness Testing Report No. HN-12-08-031

Macrostructure Report No. MA-12-08-022

Impact Testing Report No.IM-12-08-084

Welders Name Mr.Boontham J. Welder Number -

Tests Conducted By Mr.Keerati Sa-ngoen Laboratory Test LPN PLATE MILL

We certify that the statements in this record are correct and that the test coupons were prepared, welded, and tested in accordance with the requirements of section IX of the ASME Boiler and Pressure Vessel Code.

Manufacture or Contractor : QA/QC RNC

Date : 17/08/12

Inspected by : Mr.Wichit K.

Certified by : Mr.Chaloemkiat S.

Date : 17-Aug-2012

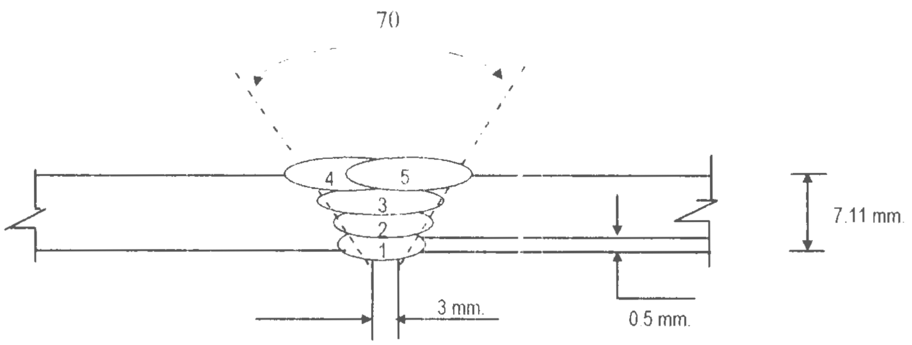


**PAE Technical Service Company Limited** Public Registration No. (5) 1571/2542

69 Srinakarin Rd., suanluang, Bangkok 10250, Thailand. Tel (662) 322-022 Fax : (622) 721-2577

| PAE TECHNICAL SERVICE | WELDING PROCEDURE | PQR No. : RNC-PQR-008 |
|---|---|---|
| | | WPS No. : RNC-WPS-008 |
| | QUALIFICATION RECORDS | THIS PQR RECORDS ACTUAL TEST |
| | | CONDITION AND RESULT |

## *Welding Sequences for RNC-PQR-008 (Butt Joint)*




6" T 7.11 mm.

| Weld Bead No. | Process | Class Filter Metal | Wire Size (mm) | Arc Current (Amps) | Arc Voltage (Volts) | Pot. AC/DC+ | String or Weave Bead | Wire Feed Speed | Rate of Travel (cm/min) | Heat Input (KJ/cm) | Gas Shielding Flow Rate (l/min) | Gas Backing Flow Rate (l/min) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | GTAW | ER316L | 2.4 | 82 | 11 | DCEN | - | - | 3.82 | 14.17 | 20 | 25 |
| 2 | GTAW | ER316L | 2.4 | 100 | 12 | DCEN | - | - | 7.92 | 9.09 | 20 | 20 |
| 3 | SMAW | E316L-16 | 3.2 | 1223 | 25 | DCEP | - | - | 6.81 | 26.88 | - | - |
| 4 | SMAW | E316L-16 | 3.2 | 121 | 25 | DCEP | - | - | 9.20 | 19.73 | - | - |
| 5 | SMAW | E316L-16 | 3.2 | 119 | 25 | DCEP | - | - | 8.91 | 20.04 | - | - |
| | | | | | | | | | | | | |
| | | | | | | | | | | | | |



*PAE Technical Service Company Limited*
69 Srinakarin Rd., suanluang, Bangkok 10250, Thailand. Tel (662) 322-022 Fax : (622) 721-2577

Public Registration No. (5) 1571/2542

**PAE TECHNICAL SERVICE CO.,LTD**
**NDT, TESTING, INSPECTION & ENGINERING CONSULTANT**
69 SOI.OONNUT 64 (SUKSAMARN) SRINAKARIN RD, SUANLUANG,BANGKOK 10250
Tel : 02) 322-0222, Fax : 02)721-2577, RAYONG (038) 607-402-3

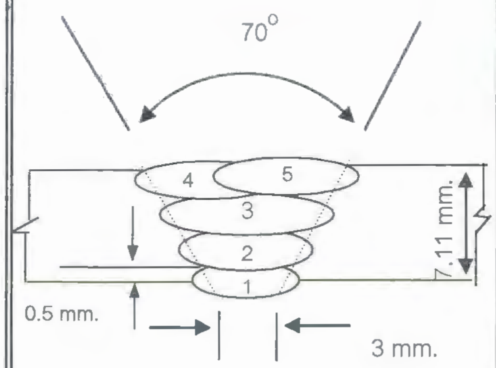
# WELDER PROCEDURE QUALIFICATION RECORD



| | | | |
|---|---|---|---|
| **Client :** Renew Coils Engineering & Supply Co.,Ltd. | **Project Name :** PQR | **Location :** RNC Work Shop | **Report No.** PQR-012 |
| | **Customer Order No. :** N/A | **WPS No. :** RNC-WPS-008 | **Date :** 6-Aug-12 |

| Welder No. | Welder Name | Welding Process | Test Position | Applicable Standard | Material Spec | Material Size | Electrode Specification | Results Visual | Results RT |
|---|---|---|---|---|---|---|---|---|---|
| - | MR.BOONTHAM J. | GTAW / SMAW | 6G | ASME - IX | A312 Gr.TP316L | ø 6" x 7.11 mm. | A 5.9 | ACCEPTED | ACCEPTED |
| | | | | | | | | | |

**Joint Details**




| Layer (S) | Process | Filler Metal Class | Filler Metal Dia. (mm.) | Current Type Polarity | Current Amp | Volt | Travel Speed cm/min | Interpas Temp °C | Heat Input (KJ / cm) | Shielding GAS | Shielding Flow Rate (l/min) | Backing GAS | Backing Flow Rate (l/min) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | GTAW | ER 316L | 2.4 | DCEN | 82 | 11 | 3.82 | 30 | 14.17 | Argon | 20 | Argon | 25 |
| 2 | GTAW | ER 316L | 2.4 | DCEN | 100 | 12 | 7.92 | 89 | 9.09 | Argon | 20 | Argon | 20 |
| 3 | SMAW | E 316L - 16 | 3.2 | DCEP | 122 | 25 | 6.81 | 99 | 26.88 | - | - | - | - |
| 4 | SMAW | E 316L - 16 | 3.2 | DCEP | 121 | 25 | 9.20 | 142 | 19.73 | - | - | - | - |
| 5 | SMAW | E 316L - 16 | 3.2 | DCEP | 119 | 25 | 8.91 | 140 | 20.04 | - | - | - | - |
| | | | | | | | | | | | | | |
| | | | | | | | | | | | | | |
| | | | | | | | | | | | | | |
| | | | | | | | | | | | | | |
| | | | | | | | | | | | | | |

| PAE Inspection By | Witness By | Approved By |
|---|---|---|
| SIGNATURE : | SIGNATURE : | SIGNATURE : |
| NAME : Mr.Wichit K. | NAME : Mr. Chatchariam S | NAME : Mr. Chatchawal S |
| DATE : 7-Aug-2012 | DATE : 4/08/12 | DATE : 7/08/12 |

PAE TECHNICAL SERVICE

PAE TECHNICAL SERVICE

# PAE TECHNICAL SERVICE CO.,LTD.

NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT

TEL : 02) 322-0222 , FAX : 02) 721-2577 , RAYONG Tel : 038)607402-3, Fax : 038)607402, SRIRACHA : 038)330220, 330229



| | | | |
|---|---|---|---|
| REPORT No. | PQR-012 | RADIOGRAPHIC EXAMINATION REPORT | INSPECTION & TEST RECORDS |
| DWG.No. | PQR | | |
| DATE : | 6-Aug-12 | | |

| | | | |
|---|---|---|---|
| CLIENT : | Renew Coils Engineering & Supply Co.,Ltd. | PROJECT NAME : | PQR |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| PROCEDURE No. : | PAE-RT-001 (Rev.0) | | EXPOSURE TIME | min. | | sec. | SOURCE TO FILM DISTANCE | 6" Inch. / - mm. | |
| RADIATION SOURCE | X-ray ( - ) MAX - kVG. γ - ray (Ir-192) | | DENSITY RANGE | 2.0 - 3.5 | | | TYPE OF PENETRAMETER | ASTM-1B | |
| | | | SENSITIVITY | 2% | | | | | |
| SOURCE RANGE | 21.10 | Ci. | FILM | TYPE | KODAK "MX" | | FILM PROCESSING | AUTO | - |
| SOURCE SIZE | 3 x 3 | mm. | | SIZE | 3.5" x 8.5" | | | MANUAL | ✓ |
| TUBE VOLTAGE | - | kVP. | INTENSIFYING SCREEN | FRONT | 0.005 | Inch. | MATERIAL | SA A312 TP316L | |
| TUBE CURRENT | - | mA | | BACK | 0.005 | Inch. | THICKNESS | 7.11 mm. | |
| APPLICABLE STANDARD : | ASME Section V Article 2 | | | | | | ACCEPTANCE CRITERIA : | ASME-IX | |

RADIOGRAPHIC TECHNIQUE : ☐ SWSI ☑ DWSI ☐ DWDI

| NAME / WQT | Position No. | Pipe/Plate Size Dia | Thk(mm). | Welder No. | Film No. | Interpretation | Results | Remarks |
|---|---|---|---|---|---|---|---|---|
| GTAW+SMAW / WPS No.RNC-PQR-008 | | | | | | | | |
| MR.BOONTHAM J. | 6G | 6" | 7.11 | - | A-B | SURFACE | ACC | |
| | | | | | B-C | NAD | ACC | |
| | | | | | C-D | NAD | ACC | |
| | | | | | D-A | P ø 1 mm. | ACC | |

Total 3.5"x8.5" 4 Films

3.5"x17" - Films

ABBREVIATIONS :

ACC : Accepted
REJ : Rejected
CP : Cluster Porosity
SF : Surface Flaw
EUC : External Udercut

WH : Worm Hole
C : Crack
P : Porosity
UC : Undercutting
SL : Slag Inclusion

LP : Lack of Penetration
LF : Lack Of Fusion
IP : Incomplete Penetration
TI : Tungsten Inclusion
BT : Burn Through

CRP : Concave Root Penetration
ERP : Excess Root Penetration
NAD : No Apparent Defects
RS : Reshoot

| | PAE INSPECTED BY | REVIEWED BY | APPROVED BY |
|---|---|---|---|
| SIGNATURE : | PAE TECHNICAL SERVICE | [signature] | |
| NAME : | Mr.Sarawut P. | Mr. Chatchawan S. | |
| DATE : | 7-Aug-12 | 7/08/12 | |

# Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnpm.co.th


23851

## Tension Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000



Report No. TS-12-08-079

Revision No. 0

| | | | |
|---|---|---|---|
| Test Product | PQR No. RNC-PQR-008 | Material Characterization | Pipe |
| Project Name | N/A | Test Method | ASTM A370-10 (Prepared by ASME IX : 2010) |
| Project No. | N/A | Standard of Test | ASME Sec. IX : 2010 |
| Material Specification | SA312 TP316L | Type of Test | Reduced Section Tension |
| Dimension | Ø 6" Sch 40 (7.11 mm Thk.) | Equipment's Capacity | 200 TONS |
| Date Received | 10 August 2012 | Equipment / Serial No. | SHIMADZU UH200A / 32098926 |
| Date Tested | 14 August 2012 | Ambient Temperature | 23 ± 5 °C |

| Item No. | Test No. | Specimen Dimension | | | | | Yield | | Ultimate | | Elongation (%) | Remark |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Diameter (mm) | Thickness (mm) | Width (mm) | Area ($mm^2$) | Gauge Length (mm) | Load (kN) | Strength ($N/mm^2$) | Load (kN) | Strength ($N/mm^2$) | | |
| 1 | 3178/12 TR1 | N/A | 6.58 | 19.51 | 128.376 | N/A | N/A | N/A | 68.803 | 535.949 | N/A | OUT OF WELD |
| 2 | 3178/12 TR2 | N/A | 6.45 | 19.46 | 125.517 | N/A | N/A | N/A | 69.196 | 551.288 | N/A | OUT OF WELD |
| | | | | | | | | | | | | |
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The unit of $N/mm^2$ is equivalent to MPa

Additional details : WPS No. RNC-WPS-008

Welding Process : GTAW+SMAW

Filler Metal/Electrode Spec't : ER-316L+E316L-16

RNC RENEW COILS ENGINEERING & SUPPLY CO., LTD. QA RNC

Reported by : ..........................

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 16 / 08 / 12

Approved by : ..........................

( Mr.Pavaret Preedawiphat )

Deputy Managing Director

Date 16 / 08 / 12

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# Testing Laboratory LPN Plate Mill Public Company Limited

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Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnpm.co.th

23856

## Bend Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000



Report No. BD-12-08-018

Revision No. 0

| | | | |
|---|---|---|---|
| Test Product | PQR No. RNC-PQR-008 | Material Characterization | Pipe |
| Project Name | N/A | Test Method | ASME Sec. IX : 2010 |
| Project No. | N/A | Standard of Test | ASME Sec. IX : 2010 |
| Material Specification | SA312 TP316L | Type of Jig | Guided-Bend Roller Jig |
| Dimension | Ø 6" Sch 40 (7.11 mm Thk.) | Diameter of Mandrel | 26 mm |
| Date Received | 10 August 2012 | Bend Angle | 180 Degree. |
| Date Tested | 14 August 2012 | Equipment / Serial No. | SHIMADZU UH200A/32098926 |

| Item No. | Test No. | Type of Bend | Size of Specimen (ThickxWidthxLength) | Indication | Location of Indication | Remark |
|---|---|---|---|---|---|---|
| 1 | 3178/12 BF1 | Transverse face bend | 6.61x37.00x163.19 mm | No Open Discontinuity | N/A | - |
| 2 | 3178/12 BF2 | Transverse face bend | 6.56x37.01x163.78 mm | No Open Discontinuity | N/A | - |
| 3 | 3178/12 BR1 | Transverse root bend | 6.58x36.95x160.84 mm | No Open Discontinuity | N/A | - |
| 4 | 3178/12 BR2 | Transverse root bend | 6.56x37.20x161.98 mm | No Open Discontinuity | N/A | - |
| | | | | | | |
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| | | | | | | |

Additional details : WPS No. RNC-WPS-008

Welding Process : GTAW+SMAW

Filler Metal/Electrode Spec't : ER-316L+E316L-16

RNC RENEW COILS ENGINEERING & SUPPLY CO., LTD.
QC/QA RNC

Reported by : ..........................................

( Mr.Keerati Sa-ngoen )

(Acting) General Manager - Quality Assurance

Date ...16.../...08.../...12...

Approved by : ..........................................

( Mr.Pavaret Preedawiphat )

Deputy Managing Director

Date ...16.../...08.../...12...

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# Testing Laboratory LPN Plate Mill Public Company Limited

12745

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Tel. 0 2815 6400-9 ext. 1179, 1210, Direct Line Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th


## Hardness Testing Report



**Customer : RENEW COILS ENGINEERING & SUPPLY CO.,LTD.**

Report No. HN-12-08-031

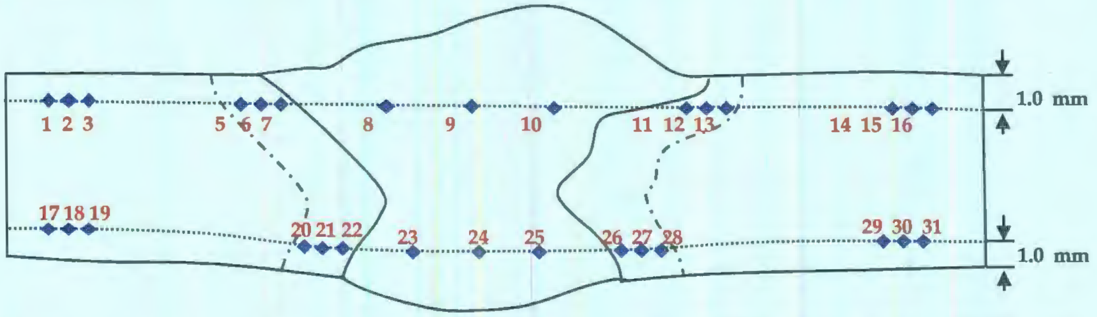
**Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.**

Revision No. 0

| | | | |
|---|---|---|---|
| Test Product | PQR No. RNC-PQR-008 | Material Characterization | Pipe |
| Test No. | 3178/12 HN | Test Method | ASTM E92-03 |
| Project Name | N/A | Standard of Test | ASTM Sec. IX : 2010 |
| Project No. | N/A | Type of test | Vicker Hardness Test |
| Material Specification | SA312 TP316L | Equipment | Mitutoyo, HV-115 |
| Dimension | Ø6" Sch 40, (7.11 mm Thk.) | Indentor Size | Diamond, pyramid 136 deg. |
| Welding Process/Position | GTAW+SMAW/All | Test Load | HV 10 Kgf. |
| Type of Joint | Butt joint | Total Force Dwell Time | 15 s |
| Date Received | 10 August 2012 | Cal. Block | No. 62137 No. A62072 |
| Date Tested | 14 August 2012 | Ambient Temperature | 25.0 °C |

Test Location

Total = 30 points




| Point No. | Location | Hardness value Value (Scale HV ) | Point No. | Location | Hardness value Value (Scale HV ) | Point No. | Location | Hardness value Value (Scale) |
|---|---|---|---|---|---|---|---|---|
| 1 | BASE | 154.0 | 16 | BASE | 158.6 | ***************************** | | |
| 2 | BASE | 153.0 | 17 | BASE | 161.4 | | | |
| 3 | BASE | 152.8 | 18 | BASE | 159.2 | | | |
| 4 | HAZ | 164.5 | 19 | HAZ | 185.6 | | | |
| 5 | HAZ | 165.4 | 20 | HAZ | 184.7 | | | |
| 6 | HAZ | 159.2 | 21 | HAZ | 186.8 | | | |
| 7 | WELD | 179.3 | 22 | WELD | 182.8 | | | |
| 8 | WELD | 182.5 | 23 | WELD | 191.4 | | | |
| 9 | WELD | 190.6 | 24 | WELD | 186.5 | | | |
| 10 | HAZ | 173.1 | 25 | HAZ | 183.1 | | | |
| 11 | HAZ | 170.6 | 26 | HAZ | 180.5 | | | |
| 12 | HAZ | 169.2 | 27 | HAZ | 182.2 | | | |
| 13 | BASE | 158.5 | 28 | BASE | 165.0 | | | |
| 14 | BASE | 156.4 | 29 | BASE | 167.7 | | | |
| 15 | BASE | 158.3 | 30 | BASE | 166.8 | | | |

**Additional details :** Filler Metal : ER-316L(GTAW) + E316L-16(SMAW)

RENEW COILS ENGINEERING & SUPPLY CO., LTD. QC/QA RNC

Reported by : ..........................

( Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurancer

Date : 15/08/12

Approved by : ..........................

( Mr.Pavaret Preedawiphat )

Deputy Managing Director

Date : 15/08/12

**Notes**

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Testing Laboratory LPN Plate Mill Public Company Limited
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Tel. 0 2815 6400-9 ext. 1179, 1210, Direct Line Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th

11623

# Macrostructure Analysis Report



Report No. MA-12-08-022

Revision No. 0

**Customer : RENEW COILS ENGINEERING & SUPPLY CO.,LTD.**

**Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.**

| | |
|---|---|
| Test Product | PQR No. RNC-PQR-008 |
| Test No. | 3178/12 MA |
| Project Name | N/A |
| Project No. | N/A |
| Material Specification | SA312 TP316L |
| Dimension | Ø6" Sch 40, (7.11 mm Thk.) |
| Welding Process/Position | GTAW+SMAW/All |
| Type of Joint | Butt joint |
| Date Received | 10 August 2012 |

| | |
|---|---|
| Material Characterization | Pipe |
| Test Method | ASME E340-00 |
| Standard of Test | ASME Sec. IX : 2010 |
| Etchant Reagent | Lepito's etch |
| Temperature | 28 °C |
| Time of Etching | 10 minutes |
| Surface Preparation | Polishing with sand paper |
| | 240, 400, 600, 800, 1000 |
| Date Tested | 11 August 2012 |


**Additional details :** The macrostructure shows complete fusion of the weldment. Weld metal and heat affected zone are free of cracks.
Filler Metal : ER-316L(GTAW) + E316L-16(SMAW)

RNC RENEW COILS ENGINEERING & SUPPLY CO., LTD. QC/QA RNC

Reported by : ..........................

( Mr.Keerati Sa-ngoen)
(Acting) General Manager - Quality Assurancer
Date : 15/08/12

Approved by : ..........................

( Mr.Pavaret Preedawiphat )
Deputy Managing Director
Date : 15/08/12

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Testing Laboratory LPN Plate Mill Public Company Limited

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Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnpm.co.th

TESTING No 0138 23861

# Impact Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000



Report No. IM-12-08-084

Revision No. 0

| | | | |
|---|---|---|---|
| Test Product | PQR No. RNC-PQR-008 | Material Characterization | Pipe |
| Project Name | N/A | Test Method | ASTM A370-10 and ASTM E23M-07a$^{\epsilon 1}$ |
| Project No. | N/A | Standard of Test | ASME Sec. IX : 2010 |
| Material Specification | SA312 TP316L | Type of Test | Charpy V-Notch Impact |
| Dimension | Ø 6" Sch 40 (7.11 mm Thk.) | Equipment / Serial No. | Tinius Olsen / 221811 |
| Date Received | 10 August 2012 | Equipment 's Capacity | 542 Joules |
| Date Tested | 15 August 2012 | Ambient Temperature | 23 ± 5 °C |

| Item No. | Test No. | Test Temperature (°C) | Size of Specimen Thickness (mm) | Width (mm) | Length (mm) | Results Energy Absorbed (Joules) | Average (Joules) | Location of Test | Remark |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 3178/12-1 | -20°C | 5.012 | 10.010 | 54.74 | 125.58 | 127.07 | Base Metal | - |
| 2 | 3178/12-2 | -20°C | 5.016 | 10.007 | 54.71 | 131.26 | | | |
| 3 | 3178/12-3 | -20°C | 5.012 | 10.006 | 54.74 | 124.37 | | | |
| 4 | 3178/12-1 | -20°C | 5.012 | 10.006 | 54.84 | 80.46 | 81.87 | Weld Metal | - |
| 5 | 3178/12-2 | -20°C | 5.008 | 9.983 | 54.87 | 81.26 | | | |
| 6 | 3178/12-3 | -20°C | 5.005 | 10.007 | 54.82 | 83.88 | | | |
| 7 | 3178/12-1 | -20°C | 5.002 | 10.004 | 54.94 | 100.16 | 103.09 | HAZ | - |
| 8 | 3178/12-2 | -20°C | 5.000 | 9.995 | 54.92 | 102.97 | | | |
| 9 | 3178/12-3 | -20°C | 5.009 | 10.002 | 54.97 | 106.15 | | | |

Additional details : WPS No. RNC-WPS-008

Welding Process : GTAW+SMAW

Filler Metal/Electrode Spec't : ER-316L+E316L-16

RNC RENEW COILS ENGINEERING & SUPPLY CO., LTD. QA RNC

Reported by : ..........................................

( Mr.Keerati Sa-ngoen )

(Acting) General Manager - Quality Assurance

Date ..16../..08../..12..

Approved by : ..........................................

( Mr.Pavaret Preedawiphat )

Deputy Managing Director

Date ..16../..08../..12..

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TKW120106 KOBELCO

# INSPECTION CERTIFICATE
## (検 査 証 明 書)

PURCHASER
需要家

WELDING ROD OR WIRE
溶加棒又はワイヤ

CERTIFICATE No.
証明書番号 : 220120244
DATE OF ISSUE
発行日 : 2012.03.21

| TRADE DESIGNATION 品名(型番) | DIMENSION 寸法(mm) | MFG.No. 製造番号 | APPLICABLE SPECIFICATION AND CLASSIFICATION 適用規格及び種類 |
|---|---|---|---|
| TG-S316L | (2.4) mm<br>3/32 inch | E36762 | AWS A5.9 ER316L<br>ASME SFA-5.9 ER316L |

CHEMICAL COMPOSITION 化学成分(%) ACCORDING TO EN10204 TYPE 3.1

| ELEMENTS 成分 | C | SI | MN | P | S | CU | NI | CR | MO | NB | N | FN | FS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ROD OR WIRE 棒又はワイヤ | 0.008 | 0.40 | 1.55 | 0.024 | 0.002 | 0.15 | 12.18 | 18.58 | 2.14 | <0.01 | 0.02 | 10 | 6 |
| ELEMENTS 成分 | FNW | | | | | | | | | | | | |
| ROD OR WIRE 棒又はワイヤ | 9 | | | | | | | | | | | | |

FS = FERRITE % (SCHAEFFLER DIAGRAM)
FN = FERRITE NUMBER (DELONG DIAGRAM)
FNW= FERRITE NUMBER (WRC)

| TENSILE PROPERTIES OF DEPOSITED METAL 溶着金属の引張性能 | | | | IMPACT PROPERTIES OF DEPOSITED METAL 溶着金属の衝撃性能 | | | HARDNESS PROPERTIES 硬さ |
|---|---|---|---|---|---|---|---|
| YIELD POINT 降伏点 | YIELD STRENGTH AT 0.2% OFFSET 0.2%耐力 | TENSILE STRENGTH 引張強さ | ELONGATION 伸び | TEST TEMP. 試験温度 | ABSORBED ENERGY 吸収エネルギー AVG.平均 | EACH | |
| (--------) N/mm² | (-) N/mm² | (---------) N/mm² | - % | (-) ℃ | (-) J | ( ) | - |
| -------- ksi | - ksi | ---------- ksi | | - ℉ | - ft-lbf | | |

| WELDING CONDITIONS 溶接条件 | | | | POSTWELD HEAT TREATMENT 溶接後熱処理 |
|---|---|---|---|---|
| TYPE OF CURRENT 電流の種類 | - | SHIELDING GAS シールドガス | - | |
| AMPERAGE 溶接電流 | - A | | | (-) ℃ × - h |
| ARC VOLTAGE アーク電圧 | - V | | | - ℉ × - h |

WE HEREBY CERTIFY THAT THE ABOVE TEST RESULTS AND MECHANICAL PROPERTIES SPECIFICATIONS SATISFY THE REQUIREMENTS OF THE APPLICABLE SPECIFICATION.
上記試験結果および機械的性質保証値は、適用規格の要求を満足していることを証明します。

KOBE STEEL, LTD.
WELDING BUSINESS FUJISAWA WORKS
株式会社 神戸製鋼所 溶接事業部門 藤沢事業所

N. Terachi

APPROVED BY CHIEF INSPECTOR
主任検査員 承認済

120321173619362119120321710 4R2191

(注) 単位、適用規格、種類を除き大文字を使用しております。
REMARKS:CAPITAL LETTERS ARE USED EXCEPT FOR UNIT AND APPLICABLE SPECIFICATION / CLASSIFICATION.